

Read the article "Driverless Cars Are Coming." Then answer the questions.

Driverless Cars Are Coming

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❷ Television and movies have long been fascinated with cars that could drive themselves. In reality, Google has had cars that could drive independently under specific conditions since 2009. Their cars have driven more than half a million miles without a crash, but so far, Google cars aren't truly driverless; they still alert the driver to take over when pulling in and out of driveways or dealing with complicated traffic issues, such as navigating through roadwork or accidents. So what roadblocks lie ahead for the autonomous car?

Sensing the World

❸ Let's begin by looking at which companies are making computer-driven cars. Originally, many futurists believed the key to developing self-driving cars someday wasn't so much smarter cars as smarter roads. For example, in the late 1950s, General Motors created a concept car that could run on a special test track. The track was embedded with an electrical cable that sent radio signals to a receiver on the front end of the car. Engineers at Berkeley tried something similar, but they used magnets with alternating polarity. The car read the positive and negative polarity as messages in binary code. These smart-road systems worked surprisingly well, but they required massive upgrades to existing roads, something that was simply too expensive to be practical.

❹ Without the option of smarter roads, manufacturers turned to smarter cars—but how much smarter did the cars need to be? For starters, they needed a whole lot of sensors. Google's modified Toyota Prius uses position-estimating sensors on the left rear wheel, a rotating sensor on the roof, a video camera mounted near the rearview mirror, four automotive radar sensors, a GPS receiver, and an inertial motion sensor. The most important bit of technology in this system is the spinning sensor on the roof. Dubbed LIDAR, it uses laser beams to form a constantly updating 3-D model of the car's surroundings. The combination of all this input is necessary for the driverless car to mimic the skill of a human at the wheel.

❺ Sensors are nothing new, of course. In the 1980s, automakers used speed sensors at the wheels in the creation of antilock brakes. Within 10 years, those sensors had become more advanced to detect and respond to the danger of out-of-control skids or rollovers. The information from the sensors can cause the car to apply brakes on individual wheels and reduce power from the engine, allowing far better response and control than a human driver could manage alone. Further improvements in sensors and computer hardware and software to make driving safer are also leading to cars that can handle more and more driving tasks on their own.

Driving or Assisting?

❻ Antilock brakes and driver assistance still seem a long way from the dream of calling a driverless cab to take us wherever we desire, but Sebastian Thrun, founder of the Google Car project, believes that the technology has finally begun to catch up to the dream. "There was no way, before 2000, to make something interesting. The sensors weren't there, the computers weren't there, and the mapping wasn't there. Radar was a device on a hilltop that cost two hundred million dollars. It wasn't something you could buy at Radio Shack." So just how driverless will the cars be in the near future?

❼ In 2013, BMW announced the development of "Traffic Jam Assistant." The car can handle driving functions at speeds up to 25 mph, but special touch sensors make sure the driver keeps hold of the wheel. In fact, none of the cars developed so far are completely driverless. They can steer, accelerate, and brake themselves, but all are designed to notify the driver when the road ahead requires human skills, such as navigating through work zones and around accidents. This means the human driver must remain alert and be ready to take over when the situation requires. This necessitates the car being ready to quickly get the driver's attention whenever a problem occurs. GM has developed driver's seats that vibrate when the vehicle is in danger of backing into an object. The Google car simply announces when the driver should be prepared to take over. Other options under consideration are flashing lights on the windshield and other heads-up displays. Manufacturers are also considering using cameras to watch that drivers are remaining focused on the road. While the driver watches the road, the car watches the driver.

❽ Why would anyone want a driverless car that still needs a driver? Wouldn't drivers get bored waiting for their turn to drive? "The psychological aspects of automation are really a challenge," admits Dr. Werner Huber, a BMW project manager driver. "We have to interpret the driving fun in a new way." Some manufacturers hope to do that by bringing in-car entertainment and information systems that use heads-up displays. Such displays can be turned off instantly when the driver needs to take over—something not available to drivers trying to text with a cell phone. In this way, the in-car system is actually a safety feature, and safety is a big concern.

Waiting on the Law

❾ Most driving laws focus on keeping drivers, passengers, and pedestrians safe, and lawmakers know that safety is best achieved with alert drivers. Presently, traffic laws are written with the assumption that the only safe car has a human driver in control at all times. As a result, in most states it is illegal even to test computer-driven cars. California, Nevada, Florida, and the District of Columbia have led the country in allowing limited use of semi-autonomous cars; manufacturers believe that more states will follow as soon as the cars are proved more reliably safe. Still, even if traffic laws change, new laws will be needed in order to cover liability in the case of an accident. If the technology fails and someone is injured, who is at fault—the driver or the manufacturer?

❿ Automakers are continuing their work on the assumption that the problems ahead will be solved. Tesla has projected a 2016 release for a car capable of driving on autopilot 90 percent of the time. Mercedes-Benz, Audi, and Nissan plan to have cars that can drive themselves by 2020. The road to the truly autonomous car stretches on ahead of us, but we grow closer to the destination every day.

Prewriting/Planning

Use the blank Prewriting/Planning space below for notes, lists, webs, outlines, or anything else that might help you plan your writing. If you need additional space for planning, raise your hand and your teacher will give you a sheet of scratch paper. **You must write your final draft in the box on the right.**

Now check your writing using this Editing Checklist.

Editing Checklist

1. Check your capitalization and punctuation.
2. Spell all words correctly.
3. Check for sentence fragments or run-on sentences.
4. Keep verb tense consistent.
5. Make sure subject and verb agree.
6. Use words according to the rules of Standard English.
7. Remember to paragraph correctly.

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Prewriting/Planning

Checklist

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In the article “Driverless Cars Are Coming,” the author presents both positive and negative aspects of driverless cars. Using details from the article, create an argument for or against the development of these cars.

Be sure to include

- your position on driverless cars
- appropriate details from the article that support your position
- an introduction, a body, and a conclusion to your argumentative essay

Final Draft

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